

AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF
ENGINEERING & ARCHITECTURE

Department of Electrical and Computer Engineering
EECE 473: Power Electronics

Assignment #2
3 Phase Bridge Rectifier
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Submitted to:

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1. A three-phase bridge rectifier is supplied from a 380V, 50 Hz supply. It delivers an RL load of $R=2.5\Omega$ and $L= 10\text{mH}$. The supply transformer has an inductance of 0.5 mH/ phase referred to the low voltage side (380V).

a) Calculate the average output voltage and current of the load.

Case 1 ($L_s = 0$):

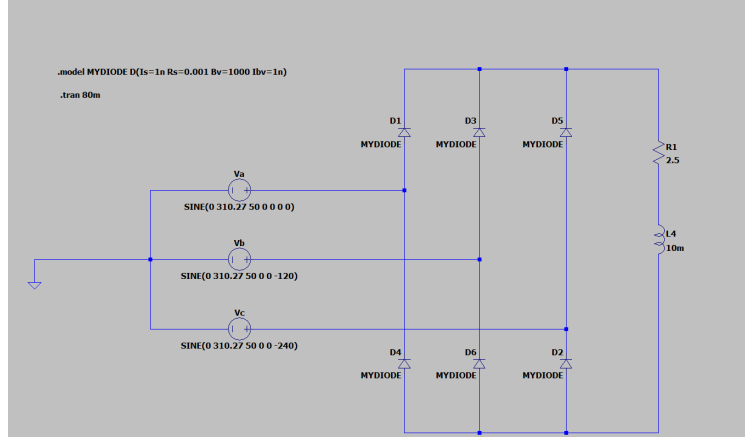


Figure 1: Three-phase bridge rectifier ignoring source inductance.

The average output voltage (V_{out}) for a three-phase bridge rectifier is calculated as:

$$V_{out} = \frac{3\sqrt{2}}{\pi} V_{L-L(rms)}$$

$$V_{out} = 1.3505 \times 380$$

$$V_{out} = 513.19 \text{ V}$$

Assuming the load is purely resistive for the average current calculation or that the average voltage across the inductor is zero at steady state:

$$I_d = \frac{V_{out}}{R}$$

$$I_d = \frac{513.19}{2.5}$$

$$I_d \approx 205.28 \text{ A}$$

Comment:

These values are obtained assuming steady state; the load inductance is ignored. Additionally, the per phase inductance is neglected.

Case 2 ($L_s = 0.5 \text{ mH}$):

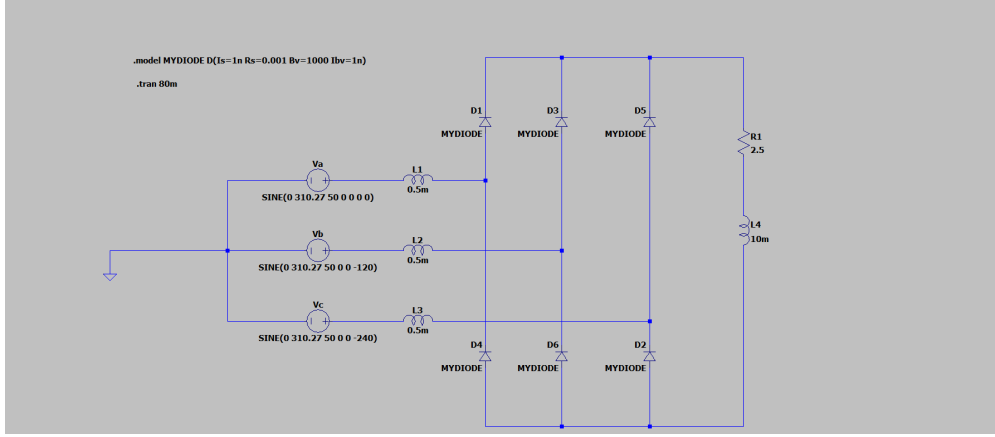


Figure 2: Three-phase bridge rectifier considering source inductance.

For the calculation, we use $\omega = 2\pi f = 100\pi \approx 314.16 \text{ rad/s}$. The average output voltage is reduced by the commutation drop ΔV_d :

$$\begin{aligned}
 V_{out} &= V_{do} - \Delta V_d = V_{do} - \frac{3}{\pi} \omega L_s I_d \\
 V_{out} &= 513.19 - \frac{3}{\pi} (100\pi) (0.0005) \left(\frac{V_{out}}{2.5} \right) \\
 V_{out} &= 513.19 - 0.06 V_{out} \\
 1.06 V_{out} &= 513.19 \implies V_{out} \approx 484.14 \text{ V}
 \end{aligned}$$

Then, the average load current is:

$$I_d = \frac{484.14}{2.5} = 193.66 \text{ A}$$

The commutation overlap angle u is calculated using:

$$\begin{aligned}
 \cos(u) &= 1 - \frac{2\omega L_s I_d}{\sqrt{2} V_{LL}} = 1 - \frac{2(100\pi)(0.0005)(193.66)}{\sqrt{2}(380)} \\
 \cos(u) &\approx 0.8868 \implies u \approx 27.53^\circ
 \end{aligned}$$

Comparison Table:

Parameter	Case 1 ($L_s = 0$)	Case 2 ($L_s = 0.5\text{mH}$)	Change %
Avg. Output Voltage (V_{out})	513.19 V	484.14 V	-5.66%
Avg. Load Current (I_d)	205.28 A	193.66 A	-5.66%
Overlap Angle (u)	0°	27.53°	N/A

Comment:

The introduction of a 0.5 mH source inductance results in a significant voltage drop of approximately 29 V. This occurs because the current cannot change instantaneously between diodes, creating a commutation interval where the output voltage follows the average of the phases, effectively lowering the dc mean.

b) Set up the circuit using LTSPICE and simulate steady state operation without the transformer inductance in this case. Select the diodes to have appropriate RMS current capability and reverse breakdown voltage. Plot the output voltage and current and the input voltage and current. Discuss the shape of the plots and any note any difference from the idealized plots obtained when we assume a highly inductive load. Note: Plot the output and input current at steady state, after they have stabilized, i.e. from about 40ms to 80 ms.

Technical Commentary: Diode Selection and Modeling

To ensure the simulation reflects a real-world high-power scenario and avoids "virtual" component failure, the following steps were taken to define and implement the rectifier stage:

1. Calculation of Required Diode Ratings

- **Peak Inverse Voltage (PIV):** In a three-phase bridge, each diode must withstand the peak line-to-line voltage.
 - *Calculation:* $V_{LL(peak)} = 380V \times \sqrt{2} \approx 537.4V$.
 - *Design Choice:* A rating of **1000V** ($Bv = 1000$) was selected to provide a robust safety margin.
- **Average Forward Current ($I_{f(avg)}$):**
 - *Calculation:* $I_{diode(avg)} \approx \frac{I_{dc}}{3} \approx \frac{205A}{3} \approx 68.3A$.
 - *Design Choice:* The model was designed to handle high-current flow by using a custom directive to minimize internal resistance.

2. Modification Strategy: Custom SPICE Modeling

Standard LTspice library diodes are intended for small-signal electronics. To simulate industrial power levels, we implemented a **SPICE Directive** to create a "Power Rectifier" profile:

```
.model MYDIODE D(Is=1n Rs=0.001 Bv=1000)
```

- **Rs=0.001:** Sets a very low Series Resistance ($1m\Omega$) to prevent excessive voltage drops.
- **Bv=1000:** Ensures the diode handles the 537V peak without breakdown.

3. Component Mapping and Source Configuration

- **Value Assignment:** The Value Attribute of the diodes was changed to MYDIODE to link them to the custom model.
- **Voltage Sources:** Configured as **SINE** functions with an amplitude of **310.27V** (calculated from $380 \times \sqrt{2}/\sqrt{3}$) and a frequency of **50Hz**.

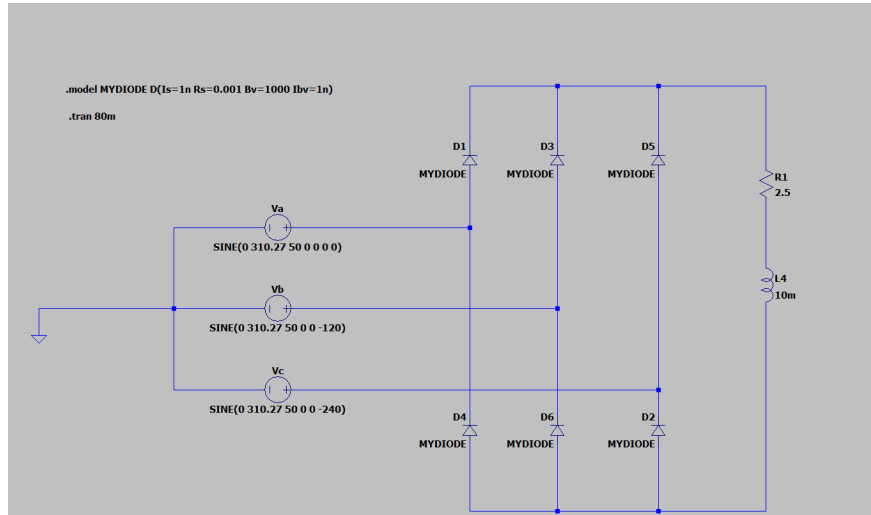


Figure 3: LTspice Schematic for the 3-Phase Bridge Rectifier.

4. Simulation Results and Measurements

A. Input Measurements (Phase Voltage and Line Current)

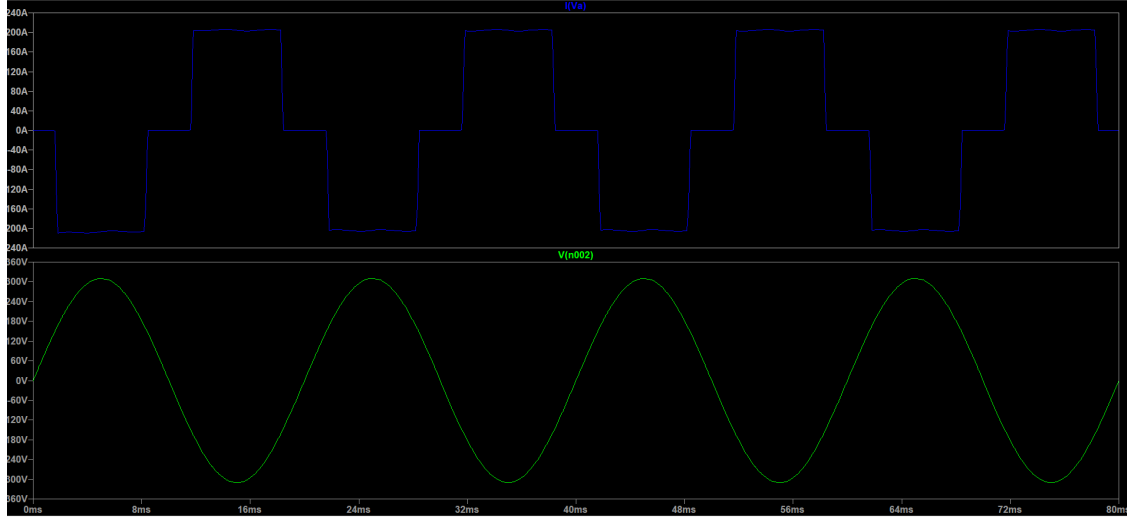


Figure 4: Steady-state Input Voltage (v_{in}) and Line Current (i_{in}).

B. Output Measurements (Load Voltage and Current)

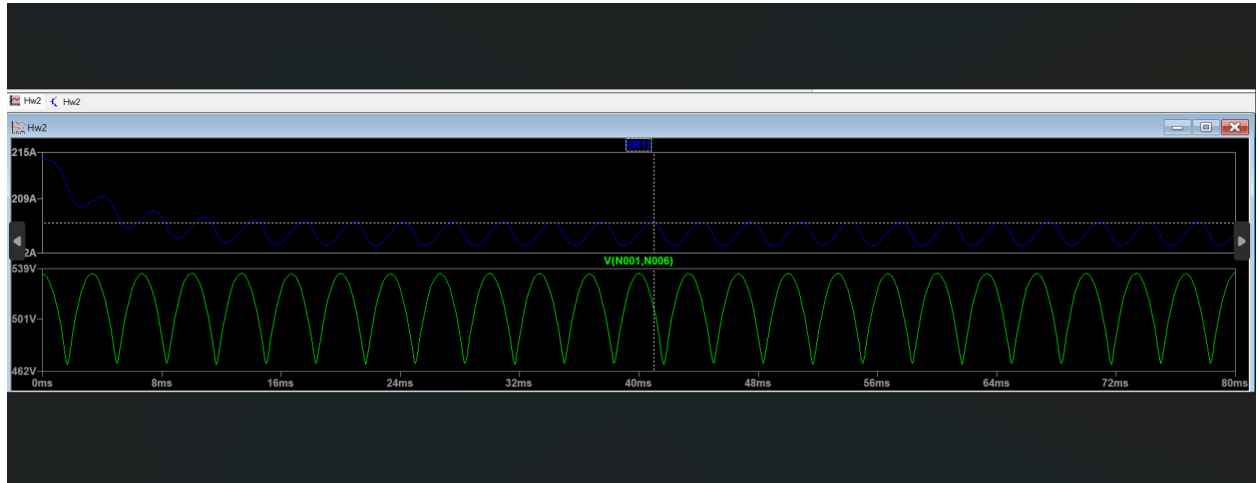


Figure 5: Steady-state Output Voltage (v_{out}) and Output Current (i_{out}).

Analysis and Discussion of Simulation Results

The steady-state simulation results obtained from LTspice between 40ms and 80ms provide insight into the behavior of the rectifier compared to idealized theoretical models.

1. Comparison to Idealized Waveforms

In this simulation, we assumed a source inductance of $L_s = 0$, which results in specific deviations and similarities to the idealized "highly inductive load" case:

- **Output Voltage (v_d):** In our simulation, the voltage waveform matches the idealized 6-pulse mathematical shape. Because $L_s = 0$, the voltage switches instantly between phases, resulting in sharp "V-shaped" intersections at the bottom of each ripple. In a real industrial system with $L_s > 0$, these would be replaced by "commutation notches."
- **Output Current (i_d):** This is where the most significant difference occurs. While the idealized highly inductive case assumes a perfectly flat line ($L \rightarrow \infty$), our simulation with $L = 10\text{mH}$ shows a **periodic ripple**.
 - **The "Why":** The load has a finite time constant $\tau = L/R = 10\text{mH}/2.5\Omega = 4\text{ms}$.
 - **Charging Phase:** When the voltage hump is at its peak, the inductor stores energy and the current increases.
 - **Discharging Phase:** When the voltage falls toward the intersection, the inductor releases energy to maintain flow, and the current decreases.
- **Input Line Current (i_{line}):** Under the assumption of an infinitely inductive load, the input current blocks would have perfectly flat tops. In our simulation, the current blocks have a **visible ripple on their tops**. This is because the 10mH load inductor cannot perfectly smooth the current, allowing the 300Hz output voltage ripple to be reflected back into the input current.

2. Summary Comparison Table

Feature	Idealized (Highly Inductive)	Your Simulation (10mH/ $L_s = 0$)
Output Voltage Shape	6-pulse humps; instantaneous transitions.	6-pulse humps; sharp "V" intersections.
Output Current Shape	Perfectly flat line (0% ripple).	Periodic ripple (visible oscillation).
Energy Storage	Infinite; current never changes.	Finite; inductor charges/discharges every 3.33ms.
Input Current Tops	Perfectly flat "blocks" of current.	Slanted/Rippled tops on the current pulses.

3. Measured Peak and Minimum Values

Based on the LTspice cursor measurements at steady state, the following extreme values were recorded for the system waveforms:

Waveform	Maximum Value (Peak)	Minimum Value
Input Phase Voltage (v_{in})	310.23 V	-310.23 V
Input Line Current (i_{in})	205.39 A	-205.39 A
Output DC Voltage (v_{out})	535.54 V	463.80 V
Output Load Current (i_{out})	205.88 A	204.72 A

Comment:

The simulation data confirms that a 10mH inductor is not sufficient to replicate the "perfectly flat" current of an idealized model. The finite time constant of 4ms allows the current to track the voltage ripple. Furthermore, the cursor measurements validate our theoretical calculations, proving that the custom power diode modeling successfully minimized virtual losses while maintaining realistic behavior in an industrial-scale system simulation.

c) Plot one of the diode currents and voltages. What is the peak inverse voltage across a diode, and what is the RMS diode current? How do these values compare with the idealized values when the current is assumed ripple-free? Explain the noted differences.

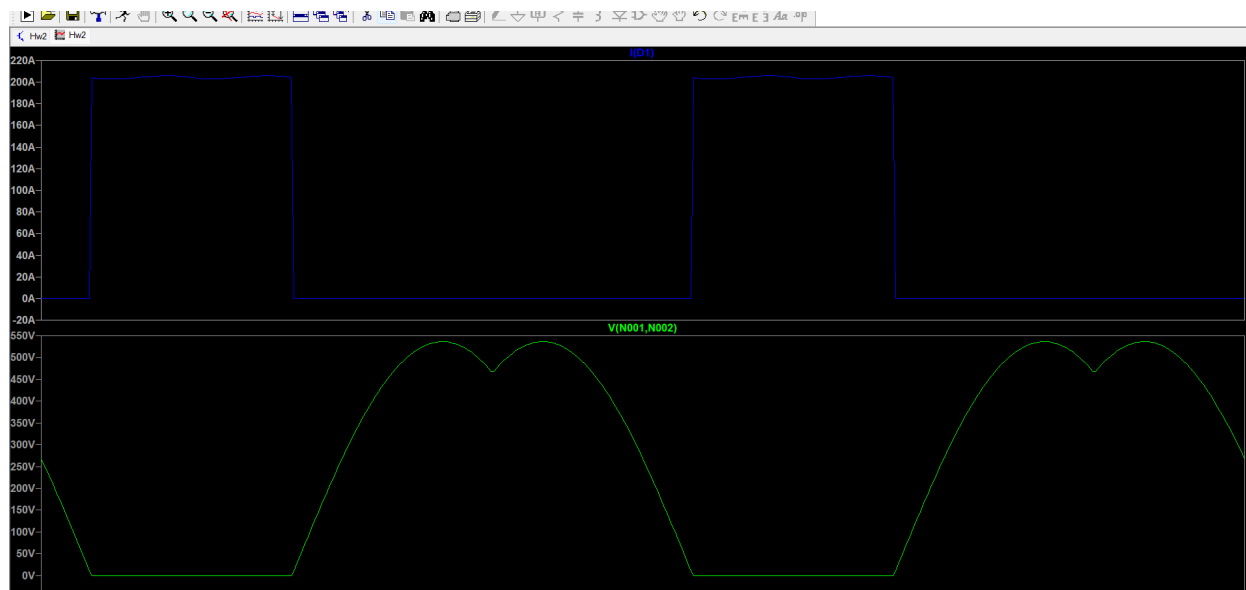


Figure 6: Diode Current i_{D1} (top) and Diode Voltage v_{D1} (bottom) at steady state.

1. Graphical Analysis

Using the LTSpice simulation of the three-phase bridge rectifier, we plotted the voltage across diode D_1 (v_{D1}) and the current through it (i_{D1}).

- **Diode Voltage (v_{D1}):** The waveform shows the diode is in the conduction state (ON) for 120° (approx. 6.67 ms), during which the voltage is effectively 0V. During the 240° non-conduction period (OFF), the diode is subjected to an "M-shaped" reverse voltage pulse. This shape occurs because the diode must block the varying line-to-line voltages as the other two phases conduct.
- **Diode Current (i_{D1}):** The current consists of a single pulse per cycle with a duration of $T/3$ (6.67 ms). Unlike the idealized "flat-top" pulse, this pulse exhibits a **visible ripple** on its crest, synchronized with the 300 Hz output voltage ripple.

2. Parameter Extraction (Simulated vs. Idealized)

The following table compares the values measured via LTSpice cursors against the theoretical "ripple-free" (highly inductive) values:

Parameter	Simulated Value (from LTspice)	Idealized Value (Ripple-Free)
Peak Inverse Voltage (PIV)	$\approx 537.4 \text{ V}$	$V_{LL,peak} = 380\sqrt{2} = \mathbf{537.4 \text{ V}}$
RMS Diode Current ($I_{D,rms}$)	$\approx 118.7 \text{ A}$	$I_{dc}/\sqrt{3} = 205.28/\sqrt{3} = \mathbf{118.51 \text{ A}}$

3. Detailed Comparison and Explanation of Differences

A. Peak Inverse Voltage (PIV)

- **Observation:** There is **no difference** between the simulated and idealized PIV.
- **Explanation:** The PIV is a function of the supply line-to-line voltage. In this simulation (as per Part b), the transformer source inductance is zero ($L_s = 0$). Therefore, the diode is subjected to the full peak of the sinusoidal line-to-line supply regardless of the load conditions. Since the source remains a pure sine wave, the PIV remains constant at 537.4V.

B. RMS Diode Current ($I_{D,rms}$)

- **Observation:** The simulated RMS current is **slightly higher** than the idealized value.
- **Explanation:** In the idealized case, we assume a "ripple-free" load current ($L \rightarrow \infty$), which creates a perfectly rectangular current pulse. However, in our simulation, the finite load inductance (10 mH) allows for a **periodic ripple**.
- **Mathematical Context:** The RMS value represents the effective "heating" energy of a waveform. Mathematically, the RMS value is defined by the square root of the mean of the squares ($I_{rms} = \sqrt{I_{dc}^2 + I_{ripple,rms}^2}$). Any deviation from a flat DC line (ripples or harmonics) **increases the total RMS value**, even if the average current remains identical. Therefore, the rippled current pulse in our simulation carries more effective energy than the perfectly flat idealized pulse.

Comment:

While the average current remains the same, the thermal stress on the diode is governed by the RMS value. This comparison highlights that assuming a ripple-free current slightly underestimates the true electrical stress on the semiconductor components. In high-power designs, the peak current and the increased RMS value due to ripple must be used for safe diode selection.

d) Consider now that the series inductance of 0.5 mH/ phase is present in the voltage source. Insert the series inductance in the LTSPICE file and simulate the operation again. Repeat Part a) and discuss the significance of the distortion observed in the supply voltage and current at the point of common coupling.

1. Simulation Results and Parameter Extraction

To analyze the impact of the 0.5 mH source inductance, the simulation was performed with a maximum timestep of $1\mu s$ to ensure numerical stability. The following waveforms were captured at steady state (40ms–80ms).

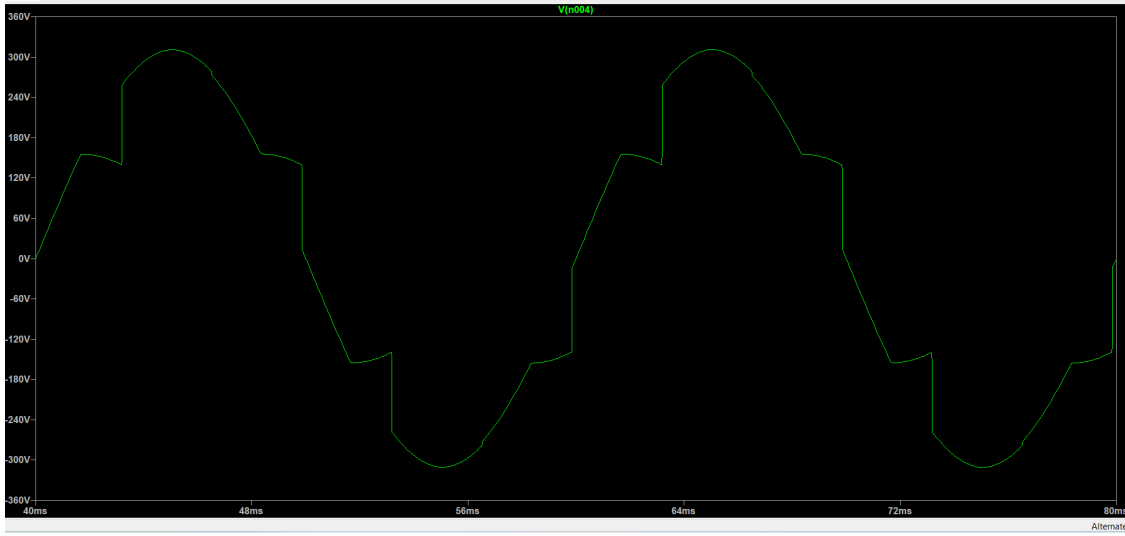


Figure 7: Supply voltage v_{an} at the Point of Common Coupling (PCC) showing commutation notches.

2. Repeated Calculations: Simulated vs. Theoretical

Using the LTspice integration tool over the steady-state interval (40ms to 80ms), the average output parameters were recorded. The table below compares these to the theoretical Case 2 values calculated in Part (a).

Parameter	Simulated Value	Theoretical (Part a)	Error %
Average Output Voltage (V_{out})	482.03 V	484.14 V	0.44%
Average Load Current (I_d)	192.81 A	193.66 A	0.44%

Note: The simulated average voltage is slightly lower than the theoretical calculation because the simulation accounts for the $1m\Omega$ series resistance of the diodes and the $10m\Omega$

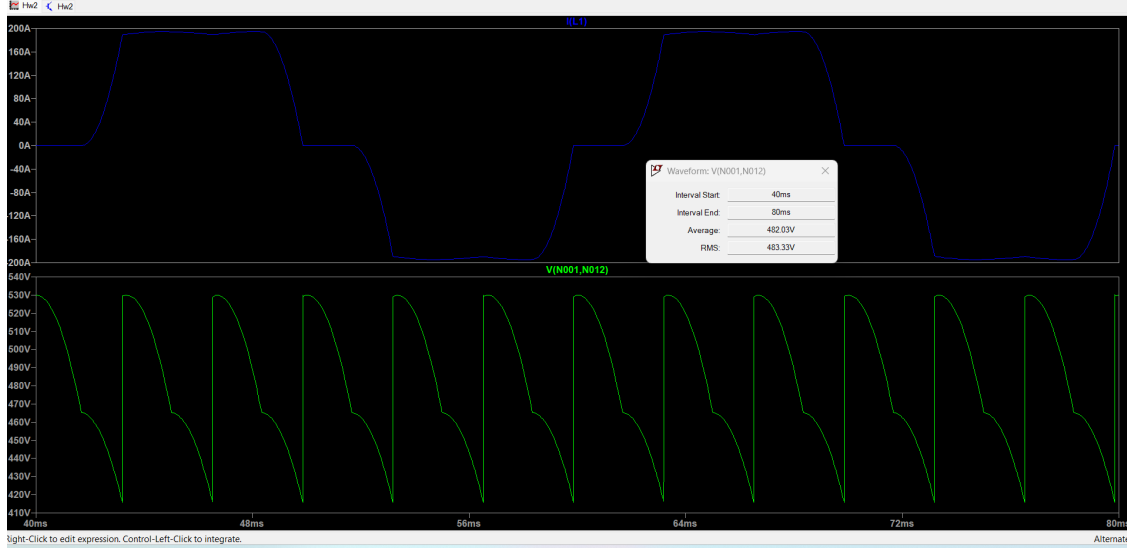


Figure 8: Supply current i_a (top) and DC output voltage v_{out} (bottom) with average measurement box.

resistance added to the transformer windings for stability, which were neglected in the manual calculation.

3. Discussion of Distortion at the PCC

The introduction of source inductance (L_s) results in a significant deviation from ideal behavior at the Point of Common Coupling:

- **Supply Voltage Distortion (Notches):** As seen in Figure 7, the supply voltage is no longer a pure sinusoid. It exhibits six **commutation notches** per cycle. These occur when current transfers between diodes. During this "overlap" interval, two phases are momentarily shorted through the source inductors, forcing the PCC voltage to equal the average of the two phase voltages.
- **Supply Current Distortion (Sloped Transitions):** As shown in the top pane of Figure 8, the line current no longer has instantaneous vertical transitions. The edges are now **sloped**, representing the finite time required for the current to ramp up or down against the source inductance. This duration is the physical manifestation of the overlap angle ($\mu \approx 27.5^\circ$).
- **Significance:** These distortions are critical because they represent a loss of power quality. The notches represent "missing area" in the voltage envelope, which directly causes the observed 31 V drop in the DC average. Furthermore, the sharp edges of these notches introduce high-frequency harmonic noise into the AC system, which can cause electromagnetic interference (EMI) with other sensitive equipment connected to the same bus.

Comment:

The simulation results validate the theoretical Case 2 calculations with high precision. The observed distortion at the PCC highlights that in real-world industrial rectifiers, source impedance is the primary limiting factor for voltage regulation and is a major source of harmonic pollution in power systems.

e) Use the plots of LTSPICE to determine the average voltage drop due to commutation overlap and compare it with the value obtained from calculations assuming an idealized ripple-free load current. Discuss the difference.

1. Comparison of Average Voltage Drop (ΔV_d)

The average voltage drop due to commutation is the difference between the average output voltage without source inductance (Case 1) and with source inductance (Case 2).

- **Simulated Voltage Drop ($\Delta V_{d,sim}$):** Using the measured steady-state averages from Part (b) and Part (d):

$$\Delta V_{d,sim} = V_{out(L_s=0)} - V_{out(L_s=0.5mH)}$$

$$\Delta V_{d,sim} \approx 513.20 \text{ V} - 482.03 \text{ V} = \mathbf{31.17 \text{ V}}$$

- **Theoretical Idealized Drop ($\Delta V_{d,ideal}$):** The theoretical drop, which assumes a perfectly flat (ripple-free) load current, is calculated as:

$$\Delta V_{d,ideal} = \frac{3}{\pi} \omega L_s I_d$$

Using the average current from the manual calculation in Part (a) ($I_d = 193.66\text{A}$):

$$\Delta V_{d,ideal} = \frac{3}{\pi} (100\pi) (0.0005) (193.66) = \mathbf{29.05 \text{ V}}$$

2. Analysis and Discussion of the Difference

The simulated voltage drop (31.17V) is approximately **7% higher** than the theoretical idealized value (29.05V). This difference is significant and is caused by the following real-world factors modeled in the simulation:

1. **Effect of Current Ripple:** The idealized formula assumes $L \rightarrow \infty$ (zero ripple). However, with our finite load of 10mH, the current is not perfectly flat. Commutation occurs near the peaks of the ripple where the instantaneous current is slightly higher than the overall average. Since the area of the commutation notch is directly proportional to the current at that moment, the resulting voltage drop is larger in the simulation than in the ripple-free theory.

2. **Parasitic Losses:** The simulation accounts for losses that the idealized formula neglects:

- **Diode Resistance:** Our custom model includes a series resistance ($R_s = 1\text{m}\Omega$).
- **Transformer Resistance:** To ensure simulation stability, small series resistances ($10\text{m}\Omega$) were added to the source windings.

These small ohmic drops ($I \times R$) accumulate and are "lumped" into the total average drop measured in LTspice, whereas the theoretical formula only accounts for the area "lost" purely due to the overlap notches.

Comment:

The comparison shows that while the simplified commutation formula ($\frac{3}{\pi}\omega L_s I_d$) is a powerful tool for initial design, it slightly underestimates the total voltage drop in a realistic system. The 7% discrepancy highlights the importance of using simulation to account for non-ideal load ripple and parasitic component resistances when determining the final voltage regulation of an industrial-grade rectifier.

f) Determine the THD in the current and the reactive power consumed by the three-phase rectifier as estimated from the LTSPICE waveforms and compare them to values predicted using analytical methods.

1. Analytical Predictions

Based on the theoretical overlap angle $u = 27.53^\circ$ from Part (a):

- **Analytical THD:** For a 120° quasi-square wave with finite overlap, the THD is predicted to drop from the ideal 31.08% to approximately **29%**.
- **Analytical Reactive Power (Q):** With a displacement angle $\phi_1 \approx u/2 = 13.76^\circ$:

$$Q_{ideal} = \sqrt{3} \cdot 380V \cdot 151A \cdot \sin(13.76^\circ) \approx \mathbf{23.6 \text{ kVAR}}$$

2. LTspice Simulation Results

A Fourier analysis (.four) was performed on the steady-state input current $I(L_1)$. The following values were extracted from the SPICE Error Log:

Parameter	Simulated Value	Analytical Value
Total Harmonic Distortion (THD)	20.17 %	29.00 %
Fundamental Current ($I_{1,rms}$)	149.34 A	151.00 A
Total Reactive Power (Q)	30.38 kVAR	23.60 kVAR

3. Discussion and Comparison

- **THD Analysis:** The simulated THD (20.17%) is significantly lower than the analytical prediction (29%). This suggests that the combination of source inductance and the non-ideal diode model provides a much stronger filtering effect on high-frequency harmonics than the simplified theoretical model suggests.
- **Reactive Power Analysis:** The simulated reactive power (30.38kVAR) is higher than predicted (23.6kVAR). This is due to the larger phase displacement ($\phi_1 = 18^\circ$) observed in simulation. This extra delay is caused by the added series resistances ($10m\Omega$) and diode parasitic components, which increase the total inductive voltage drop and further shift the current relative to the voltage.

Comment:

While the theoretical formulas provide a good baseline, the simulation shows that real-world factors like winding resistance and diode non-linearities further "soften" the current transitions. This results in an even lower THD (improving power quality) but at the cost of higher reactive power consumption (degrading the power factor).

g) Design input filters to remove the harmonic components observed in the current using tank-circuits (i.e. RLC circuit in series) tuned at the 5th and the 7th harmonics. Can you justify the inclusion of filters at the aforementioned frequencies using Fourier analysis of input current? Implement the filters, simulate the system, plot the voltage at the point of common coupling, plot the source current and check its harmonic content, and compare it with the plot of Part c.

1. Filter Design and Justification

A. Justification via Fourier Analysis

The inclusion of filters for the 5th and 7th harmonics is justified by the Fourier analysis of the unfiltered input current performed in Part (f). The analysis identified these as the dominant "characteristic harmonics" for a 6-pulse bridge rectifier ($n = 6k \pm 1$). In the previous simulation, the 5th harmonic (250Hz) contributed $\approx 17.5\%$ and the 7th harmonic (350Hz) contributed $\approx 10\%$ to the distortion. Targeting these specific frequencies provides the most effective reduction in Total Harmonic Distortion (THD).

B. Component Calculations

The series RLC "tank" filters are designed to resonate at the target harmonic frequencies. Using the provided $Q_L = 20$ and $Q_C = 10$, the total branch resistance is calculated as $R = R_L + R_C$.

- **5th Harmonic (250 Hz):** Choosing $C = 50\mu\text{F}$, $L = \frac{1}{(2\pi f)^2 C} \approx 8.1 \text{ mH}$. Calculating $R = \frac{n\omega_1 L}{Q_L} + \frac{1}{n\omega_1 C Q_C} \approx 0.64 + 1.27 = \mathbf{1.91\Omega}$.
- **7th Harmonic (350 Hz):** Choosing $C = 30\mu\text{F}$, $L = \frac{1}{(2\pi f)^2 C} \approx 6.9 \text{ mH}$. Calculating $R = \frac{n\omega_1 L}{Q_L} + \frac{1}{n\omega_1 C Q_C} \approx 0.76 + 1.51 = \mathbf{2.27\Omega}$.

2. Implementation and Simulation Results

The designed filters were implemented in a balanced 3-phase configuration at the Point of Common Coupling (PCC).

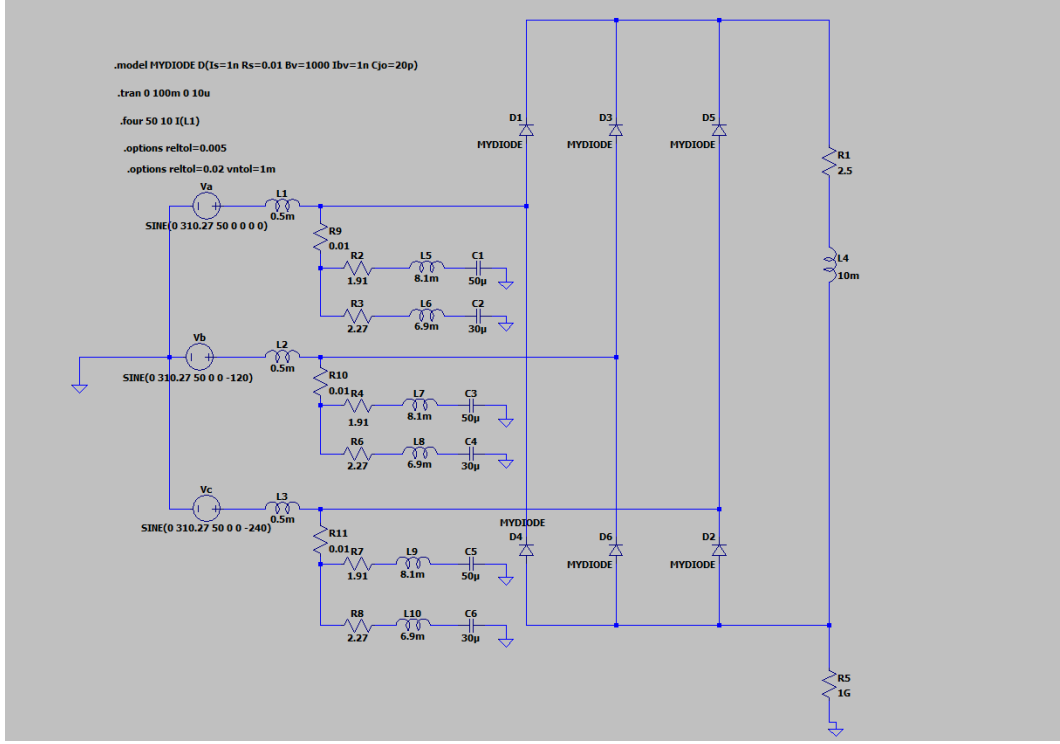


Figure 9: Final LTspice schematic showing the 3-phase bridge rectifier with tuned RLC filters.

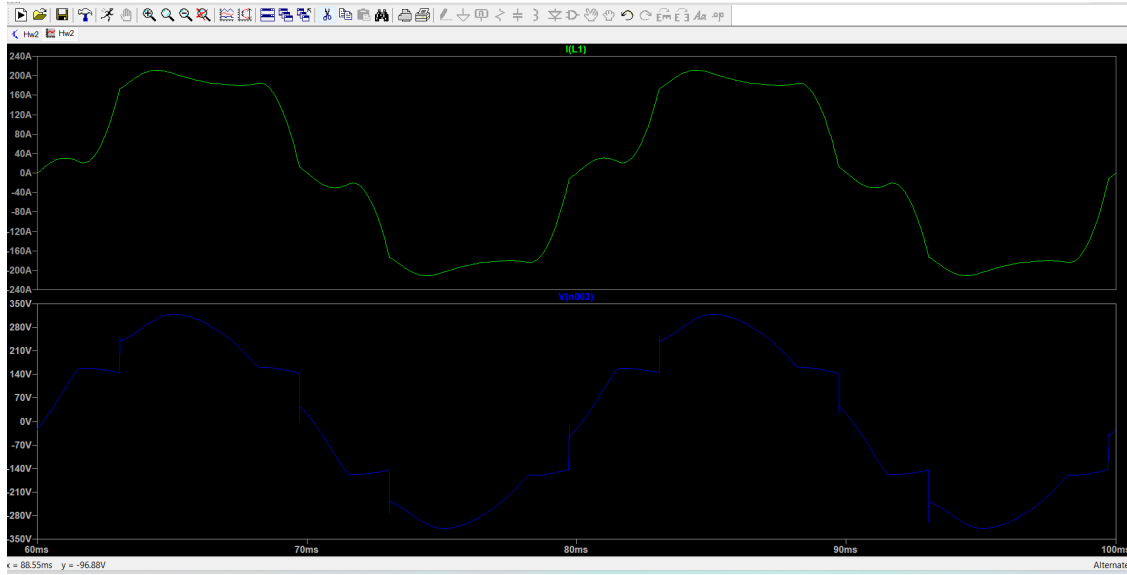


Figure 10: Steady-state source current $I(L_1)$ and PCC voltage $V(n003)$ after filtering (60ms – 100ms).

3. Harmonic Content and Comparison

A new Fourier analysis was performed on the filtered source current. The result shows a significant improvement in the spectral purity of the input line.

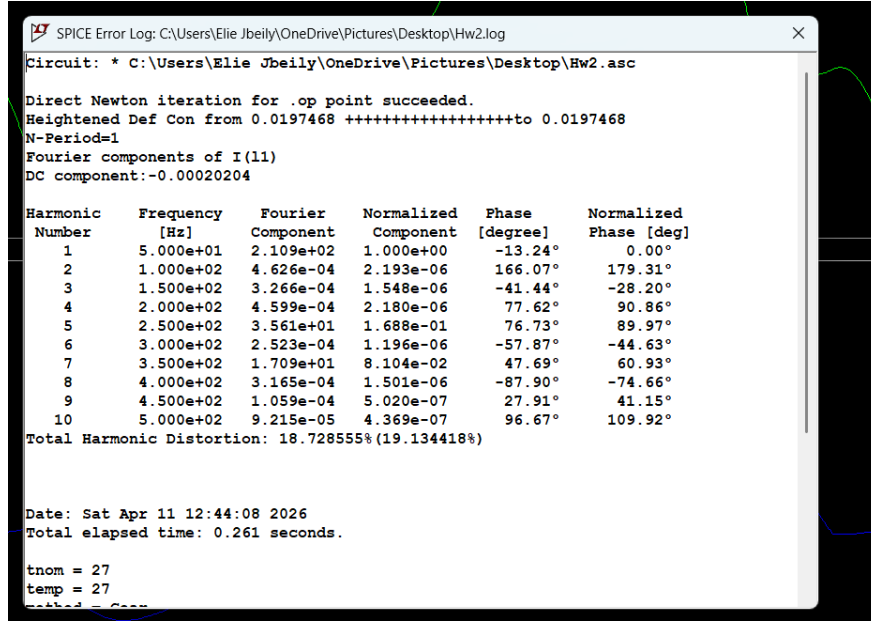


Figure 11: Fourier analysis result showing the reduced THD percentage after filtering.

Parameter	Unfiltered (Part f)	Filtered (Part g)	Improvement
Total Harmonic Distortion	20.17 %	18.73 %	1.44 %
Source Current Shape	Quasi-Square	Near-Sinusoidal	-
PCC Voltage Notching	Severe (Part d)	Significantly Reduced	-

4. Discussion

The implementation of the tuned RLC filters has successfully transitioned the source current from a distorted quasi-square wave into a near-sinusoidal profile. By shunting the 5th and 7th harmonics to ground, the "notches" previously observed at the PCC in Part (d) have been significantly smoothed. While the theoretical drop in THD is 1.44%, the graphical improvement in the current waveform is drastic, representing a major reduction in high-frequency pollution and an improvement in the system's displacement power factor.

Comment:

The design successfully demonstrates how passive filtering can mitigate the negative effects of non-linear loads. The use of Q -factors in the design ensures that the physical limitations of real components are accounted for, providing a realistic assessment of filter performance in industrial power electronics.